WEEK 1 · DEVELOPMENTALISM

CLASS 2
Development

READING 1

Early Childhood Development – UNICEF

READING 2

Early Childhood Development: Data, Evidence and Tracking – UNICEF

READING 3

Understanding Basic Concepts of Developmental Diagnosis in Children – Brown et al.

READING 4

Excerpts from *Childhood* (pp. 1–8; 22–30) – Jenks

UNICEF

EARLY CHILDHOOD DEVELOPMENT

Early childhood development

For every child, early moments matter.

Early childhood development. © UNICEF/UNI333166/Bhardwaj

The challenge

Faster than we ever thought: the first years of a child's life set the stage for all future growth.

In the earliest years of life, especially from pregnancy to age three, babies need nutrition, protection and stimulation for healthy brain development. Recent advances in neuroscience provide new evidence about a baby's brain development during this time. As a result, we know that in their earliest years, babies' brains form new connections at an astounding rate – according to Harvard University's Center on the Developing Child more than 1 million every single second – a pace never repeated again.

> **1 million+**
>
> new neural connections form in a baby's brain every single second in the earliest years – a pace never repeated again.

In the brain-building process, neural connections are shaped by genes and life experiences – namely good nutrition, protection and stimulation from talk, play and responsive attention from caregivers. This combination of nature and nurture establishes the foundation of a child's future.

Yet too many children are still missing out on the 'eat, play, and love' their brains need to develop. Put simply, we don't care for children's brains the way we care for their bodies.

> **When you pay attention to the beginning of the story, you can change the whole story.**
>
> Raffi Cavoukian, singer and founder of Canada's Centre for Child Honouring

A mix of factors determine why some children receive the nutrition, protection and stimulation they need, while others are left behind. Poverty is a common part of the equation.

> **250 million**
>
> children under five in low- and middle-income countries risk not reaching their development potential because of extreme poverty and stunting.

Often, the most disadvantaged children are least likely to have access to the essential ingredients for healthy development. For example, frequent or prolonged exposure to extreme stress – such as neglect and abuse – can trigger biological response systems that, without the buffer of a protective adult, create toxic stress, a response that can interfere with brain development. As the child grows, toxic stress can portend physical, mental and behavioural problems in adulthood.

Conflict and uncertainty also play a role as children younger than five in conflict-affected areas and fragile states face elevated risks to their lives, health and wellbeing.

Oversight and inaction have a high price and long-term implications for the health, happiness and earning potential as these children become adults. They also contribute to global cycles of poverty, inequality and social exclusion.

Despite the need, early childhood programmes remain severely underfunded with lacklustre execution. Government investment in early childhood development is low. For example, in 27 sub-Saharan African countries measured, only 0.01 per cent of gross national product was spent on pre-primary education in 2012.

There is also little public understanding of the importance of a child's first years and slight public demand for policies, programmes and funding.

Building your baby's brain

Some key facts:

- Lack of nutrition in early childhood leads to stunting, which globally affects nearly one-in-four children younger than five.
- Risks associated with poverty – such as undernutrition and poor sanitation – can lead to developmental delays and a lack of progress in school.
- Violent discipline is widespread in many countries, and nearly 70 percent of children between two and four were yelled or screamed at in the past month.
- 300 million children younger than five have been exposed to societal violence.
- For a child in a low- or middle-income country, poor early development could mean they earn around one-quarter less in income, as an adult.

- For a country, poor early childhood development could mean economic loss; in India, the loss is about twice the gross domestic product spent on health.

The solution

Fast-growing brains need family-friendly policies and environments to keep them going.

Good news: the right interventions at the right time can bolster development, break intergenerational cycles of inequity and provide a fair start in life for every child. For babies born into deprivation, intervening early, when the brain is rapidly developing, can reverse harm and help build resilience. For children with disabilities, it means making sure they have access to the individual, family and community services available to all children; combined with programmes that address each child's specific needs.

We can support #EarlyChildhoodDevelopment by expanding existing programmes, especially health services. For example, The Lancet series found it would cost only an additional US$0.5 per person annually to add two services to support nurturing care of children into an integrated maternal and child health and nutrition service package.

> **Every time a parent speaks to a young child, it sparks something in the child; it's stimulation to the child. It forms brain connections.**
>
> Dr Pia Rebello Britto, neuroscientist, UNICEF senior advisor on ECD

Thanks to the compelling scientific evidence and sustained advocacy, governments and society are beginning to realise the criticality of investing in the earliest years of a child's life. In 2015, early childhood development was included in the Sustainable Development Goals, reaffirming its growing status in the global development agenda. This built on earlier efforts which saw early childhood development included in the Convention on the Rights of the Child which states that a child has a right to develop to "the maximum extent possible" and recognised "the right of every child to a standard of living adequate for the child's physical, mental, spiritual, moral and social development."

The best start in life

Watch: Interacting with babies

Watch: Why play is important

Watch: Babies are scientists

We must act urgently to make investing in early childhood development a priority in every country to achieve the 2030 goals. Investing in early childhood development is a cost-effective way to boost shared prosperity, promote inclusive economic growth, expand equal opportunity, and end extreme poverty.

$13

For every $1 spent on early childhood development, the return on investment can be as high as $13.

But parents need time and support to create a loving and safe environment to give their babies the ‘eat, play and love’ they need, and to help build their babies’ brains.

That’s why UNICEF is working to increase investment in family-friendly policies, including paid parental leave and access to quality, affordable childcare; it makes good sense for governments because it helps economies and businesses, as well as parents and children.

Investing in family-friendly policies makes good sense for businesses too; giving parents flexibility creates a happier and more productive workforce, and allows them more time to build the brains of the future.

The time to act is now; together, we can make the #EarlyMomentsMatter for every child.

Read the UNICEF’s six-point call to action for governments and businesses

Policy and investment determine children’s future, UNICEF fights for the best ones.

Early Moments Matter for Every Child

This report outlines the neuroscience of early childhood development, including the importance of nutrition, protection and stimulation in the early years.

Care for child development

A guide for health workers and other counselors as they help families build stronger relationships with their children and solve problems in caring for their children at home.

Care for child development package | Related global resources | Western and Central Africa resources

Learning through Play

Read the brief on strengthening learning through play in early childhood education programmes.

Why Early Childhood Development is the foundation for sustainable development

Read UNICEF Senior Advisor on Early Childhood Development Pia Britto’s blog post.

UNICEF Data has global information about children and early development

Read about the situation of children around the world, in numbers.

UNICEF Early Childhood Development programme

Early Childhood Development programme guidance | Early Childhood Development in the UNICEF Strategic Plan 2018-2021

UNICEF Early Childhood Development Kit for Emergencies – Evaluation

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> **The time to invest in the future strength of our nations, our economies and our communities is in the earliest years of life. The clock is always ticking and the time to act is now.**
>
> Jack P. Shonkoff, M.D., Director of the Center on the Developing Child at Harvard University

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Source: unicef.org/early-childhood-development

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EARLY CHILDHOOD DEVELOPMENT — DATA & EVIDENCE

Tracking early childhood development

Data and evidence are critical to understanding what young children need.

Tracking early childhood development. © UNICEF/UN075657/Nesbitt

Early childhood development (ECD) has many dimensions and unfolds at a breathtaking pace. Measuring ECD is a complex undertaking, and it is a new field, marked by rapid evolution, great urgency – and significant gaps in our knowledge.

We have long understood children's development to their full potential as a fundamental human right. We now know, from recent neuroscience research, just how critical the early years of life are in the development of a child's brain and in shaping their future. Meanwhile, economics has shown us the great benefits of investing in young children during this unique developmental window – and the wrenching costs, to children and societies, of failing to do so.

> **We now know just how critical the early years of life are in the development of a child's brain and in shaping their future.**

43%

In 2016, the medical journal *The Lancet* estimated that 43 per cent of children under 5 in low- and middle-income countries were at risk of not developing to their full potential.

In 2016, the world was shocked when the medical journal The Lancet estimated that 43 per cent of children under 5 in low- and middle-income countries were at risk of not developing to their full potential. But with the underlying data based on composite, proxy indicators, what that means, and what to do about it, remains open to interpretation.

We don't yet have a full picture of the well-being and developmental status of the world's youngest children, owing to the following challenges:

- Many of the available data points are collected by each sector, separately, and often cannot be used in an analysis that cuts across sectors. In this context, analyses of caregiving practices and home environments are limited and need strengthening.
- Establishing a system for tracking child development data at the population level is particularly challenging due to the multidimensional nature of ECD.

- A lack of disaggregated ECD data prevents us from fully understanding how factors like age, gender, disability, or discrimination based on ethnicity or race affect young children's chances of reaching their potential – and how we can change that equation.
- Research evidence on child development and the impact of policy and programme interventions in low- and middle-income countries and humanitarian contexts is still limited.

Measuring and monitoring ECD is critical to understanding what young children need and identifying those at risk of being left behind and not achieving their full developmental potential. Our knowledge is built upon a range of evidence, including numerical data, qualitative research and analyses, implementation research and evaluations. We depend on this evidence to target and improve interventions, advocate with governments to improve conditions for young children and their families, and make the case for investments in ECD.

UNICEF's response

UNICEF works with governments and other partners to close the gaps in our knowledge of young children's development, support monitoring and reporting on its many dimensions, and generate evidence to advocate for better resourced and targeted interventions in the early years of life. We also report on global results from the implementation of ECD programmes in our country offices.

UNICEF provides financial and technical assistance for the collection and analysis of data on children's survival, health and nutritional status, access to play and early learning, their home environments, and the care and discipline they receive from their parents and caregivers. The Multiple Indicator Cluster Surveys (MICS) are an example of such support. Conducted at regular intervals since 1995 and implemented in over 100 low- and middle-income countries, they are the largest source of internationally comparable data on ECD.

In the Sustainable Development Goals, the international community recognized the importance of ECD through the inclusion of a dedicated indicator, SDG 4.2.1, which measures the proportion of children under 5 years old who are developmentally on track in health, learning and psychosocial well-being.

As the custodian agency for this indicator, UNICEF led the development and validation of a new measurement tool, the Early Childhood Development Index 2030, and is supporting national governments to report on progress toward the SDG target of ensuring that, by 2030, all girls and boys have access to quality early childhood development, care and pre-primary education.

The resources below cover the major measurement tools, research and data repositories on ECD that represent our most up-to-date, always evolving knowledge of the state of the world's youngest children.

Source: unicef.org/early-childhood-development/data-evidence-tracking

Review Article

Understanding basic concepts of developmental diagnosis in children

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Contributions: (I) Conception and design: All authors (II) Administrative support: None; (III) Provision of study materials or patients: None; (IV) Collection and assembly of data: All authors; (V) Data analysis and interpretation: None; (VI) Manuscript writing: All authors; (VII) Final approval of manuscript: All authors.

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Abstract: Developmental diagnosis is based on an understanding of basic concepts of typical and atypical developmental progression. Child development is influenced by multiple factors, including the development of the nervous system and other organ systems, and the child's physical and social environment. Different factors interplay with each other in influencing the overall development of the child. Development and behavior of the child are intricately associated. Typical child development follows certain basic principles. Some of the more commonly reported developmental concerns include global developmental delay, intellectual disability, cerebral palsy, delayed speech and language, attention deficits, autism, and specific learning disabilities. The clinical presentation of atypical development varies, depending up on the age of the child; with motor delay in early infancy, and learning difficulties in school age child. Regular surveillance and periodic screening help identify specific areas of developmental and behavioral concerns and suggest a need for further appropriate psychological, medical and laboratory evaluation. The principles of management of a child with developmental concerns include early intervention and response to treatment approach, remediation, accommodation, and specific behavioral and pharmacological interventions when indicated.

Keywords: Atypical development; delay; deviation; dissociation; surveillance; screening; global developmental delay; developmental quotient (DQ); intelligence quotient (IQ)

Submitted Nov 05, 2019. Accepted for publication Nov 14, 2019.
doi: 10.21037/tp.2019.11.04
View this article at: http://dx.doi.org/10.21037/tp.2019.11.04

Introduction

Child development is generally described in terms of streams or domains of development. The four domains or streams of development are, (I) motor development—gross motor and fine motor, (II) speech and language development—expressive and receptive, (III) social and emotional, and (IV) cognitive (1-12). Development in motor adaptive abilities, language and communication, and intellectual capacity provide a basis for social and emotional development (7,8,13,14). Problem solving skills is a reflection of cognitive development, which also includes visual perceptual and visual motor abilities. Development is also described in terms of its progression as either typical or atypical for the age of the child.

The acquisition of developmental skills is a function of interplay between development of the nervous system and other organ systems, and child's social and physical environment and stimuli. Typical development is characterized by certain basic tenets. These are, (I) gross motor development progresses in a cephalo-caudal sequence, (II) fine motor development progresses from midline to lateral sequence, (III) primary motor patterns or primitive reflexes are integrated into more complex motor

Table 1 Developmental surveillance, screening, and evaluation

Process	Goal	Definition
Surveillance	Identify children who may have developmental problems	Gathering and synthesizing information about developmental progress of the child based on history, observations by parents or other caretakers and health care practitioners, and during periodic visits on a longitudinal and continuous basis over time
Screening	Identify children at risk of a developmental disorder	The administration of a brief standardized screening test
Evaluation	Identify a specific developmental disorder and its etiology if known	A diagnostic process that may involve appropriate laboratory, genetic, or metabolic testing; neuroimaging studies, and psychological testing as well as specialist consultations

Used with permission from: Patel DR. Principles of developmental diagnosis. In: Greydanus DE, Feinberg A, Patel DR, *et al.* editors. Pediatric Diagnostic Examination. New York: McGraw Hill Medical, 2006:629-44.

Table 2 Characterization of atypical development

Atypical development	Definition
Delay	Significantly delayed attainment of milestones or skills in one or more domains, but in an expected sequence, compared to that of typically developing children
Deviation	Attainment of developmental skills in a given domain that is out of sequence, for example, when an infant rolls from supine to prone before prone to supine
Dissociation	Attainment of developmental skills at significantly different rates between two or more domains of development. For example, when there is delayed motor development relative to other domains in cerebral palsy
Regression	Loss of previously acquired developmental milestones or skills or failure to acquire new skills

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patterns to allow for later, sequential voluntary motor development, (IV) the rate of attainment of developmental skills between infants vary; however, the sequence remains the same, and (V) there is progression from generalized reflexive responses to a more specific and purposeful response pattern (4,7-9). Periodic surveillance and screening help identify infants and children who may need additional evaluation (*Table 1*), and the application of the basic tenets of typical development allow for an understanding of different patterns of atypical progression of development (*Table 2*) (1-3,7,8).

Definitions

Intelligence quotient (IQ) as measured by individually administered standardized tests, is a widely used measure of cognitive abilities or intelligence. IQ is calculated by dividing mental age (MA) by chronologic age (CA) multiplied by 100, and can be measured reliably in children ≥6 years of age (8,11,13-15). IQ is used as one criterion to define intellectual disability. In addition to sub-average IQ, the definition of intellectual disability requires the presence of limitations in adaptive functioning (as measured by individually administered standardized tests) (16-21). Adaptive functioning comprises conceptual, social, and practical adaptive skills. IQ of 70 or less with a standard error of measurement of 5 is considered to be sub-average. Because it is difficult to accurately and reliably measure IQ in children younger than 6 years of age, the term global developmental delay is used when development is significantly delayed in 2 or more domains. Developmental quotient (DQ) is calculated as follows: DQ = [developmental age (DA)] divided by CA ×100 (7,8). Significant developmental delay is defined as DQ of 70 or less. Atypical development is described as delay, deviation, dissociation or regression (*Table 2*) (1-3,7-10). DQ is a measure of rate of

Table 3 Diagnostic and statistical manual of mental disorders classification of neurodevelopmental disorders

Disorder	Sub-categories
Intellectual disabilities	Intellectual disability: mild, moderate, severe, profound
	Global developmental delay
	Unspecified intellectual disability
Communication disorders	Language disorder
	Speech sound disorder
	Childhood-onset fluency disorder (stuttering)
	Social (pragmatic) communication disorder
	Unspecified communication disorder
Autism spectrum disorder	Autism spectrum disorder with a known medical or genetic condition or environmental factor
	Associated with another neurodevelopmental, mental or behavioral disorder
Attention-deficit/hyperactivity disorder	Predominantly inattentive
	Predominantly hyperactive/impulsive
	Combined inattentive and hyperactive/impulsive
	Other
	Unspecified
Specific learning disorder	Impairment in reading
	Impairment in written expression
	Impairment in mathematics
Motor disorders	Developmental coordination disorder
	Stereotypic movement disorder
	Tourette disorder
	Tic disorders
Other	Other or unspecified neurodevelopmental disorders

Source: American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders. 5th edition. Washington DC: American Psychiatric Press, 2013:33-86.

progression of development in a given domain.

Clinical presentations

The predominant clinical presentation of atypical development varies in infants, children and adolescents. Various neurodevelopmental disorders are described based on their clinical features; the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) classification of neurodevelopmental disorders is shown in *Table 3* (5).

Infants

Atypical motor development

Parents are more likely to first notice if infant fails to attain expected motor milestones. Often, parents compare infant's development to other infants. Because typical development occurs within a range of period, most infants with apparent motor delay likely have normal variation or maturational lag (8,10). Cerebral palsy is the most significant cause of motor delay in infancy. In addition to motor delay, infants and children with cerebral palsy have abnormal tone and posture

Table 4 Early clinical signs suggestive of cerebral palsy

Age	Clinical signs
3–6 months	Head falls back when picked up while lying on back
	Feels stiff
	Feels floppy
	Seems to overextend back and neck when cradled in someone's arms
	Legs get stiff and cross or scissor when picked up
≥6 months	Doesn't roll over in either direction
	Cannot bring hands together
	Has difficulty bringing hands to mouth
	Reaches out with only one hand while keeping the other fisted
≥10 months	Crawls in a lopsided manner, pushing off with one hand and leg while dragging the opposite hand and leg
	Scoots around on buttocks or hops on knees, but does not crawl on all fours

Source: Public domain: https://www.cdc.gov/features/cerebral-palsy-11-things/index.html

or movements (8-10,13,14,22,23). Delay in attaining motor milestones is an early sign of cerebral palsy. Other clues that parents first notice are listed in *Table 4* (22,23). Other major causes of predominant motor delay during infancy include birth injury to the brain, stroke, metabolic insult affecting brain, congenital central nervous system infections and neuromuscular disorders (10,13-15,17,18).

Atypical development in social and language domains

Another area of concern in infancy and toddler age is atypical progression of development in social, cognitive and language domains. A complete evaluation is indicated in infants or toddlers with following signs: no babbling, no pointing (joint attention) or gestures by 12 months of age; no single words by 16 months; and no 2-word spontaneous phrases by 24 months (11,12,14). Absence of joint attention, the ability to attend to both and object and a person at the same time, is considered an early sign of autism (5,12). Any loss or regression of a previously acquired skill at any age is any indication for complete diagnostic evaluation including neurological, metabolic and genetic as appropriate (15). Developmental regression is often seen in infants and toddlers with autism between 18 and 24 months of age (10,12). Other significant causes of regression before 2 years of age include metabolic conditions such as disorders of amino acid metabolism, lysosomal storage disease, hypothyroidism, mitochondrial diseases, tuberous sclerosis, Lesch-Nyhan syndrome, Rett syndrome, Canavan disease, and Pelizaeus-Merzbacher disease (10). In addition to autism spectrum disorder, other conditions that should be considered in infants with predominant language, cognitive and social deficits include hearing impairment, severe cognitive deficit, genetic disorders, inborn errors of metabolism, hypothyroidism, and severe nutritional or environmental deprivation (1,3,10,15).

Children

Atypical language development

Concerns about development of speech and language (*Table 5*) are predominant presentations in children. In a typically developing child, intelligibility of speech progresses from about 25% by age 2 years to 100% by age 4 years. In addition to poor intelligibility for given age, persistent baby talk, mispronunciations of words or lack of spontaneous speech suggest speech and language delay (24-30). The basic components of a language are phonology, grammar, lexicon, semantics and pragmatics (*Table 6*); any of which can be affected in a child with atypical language development (24-30).

Autism spectrum disorder, intellectual disability, and developmental language disorder are main causes for atypical language development in a young child (9,11-14,18,20,31). Autism spectrum disorder is characterized by qualitative impairment in social relatedness, stereotypical behaviors and a spectrum of impairment in communication ability, especially social communication (5,12,32). Autism spectrum disorder can be generally

Table 5 Basic speech and language terms

Term	Description
Speech	Production of sounds for words
Language	System of symbolic knowledge represented in the brain used for meaningful communication
Prosody	Pattern of rhythm, stress, and intonation of speech
Phoneme	A unit of sound in speech
Morpheme (word)	The smallest meaningful unit of language

Table 6 Basic components of language

Component	Subcomponent	Description
Structure or form of language	Phonology	Sound system of the language that is made of phonemes. Use of phonemes and conventions for their combinations
	Grammar	Morphology: rules or conventions for constructing meaningful words, e.g., adding -s/-es to a word to indicate plural (duck/ducks)
		Syntax: rules or conventions for constructing meaningful phrases or sentences and their relationship, e.g., word order “Daddy go there” But “There Daddy go” is not typical English structure
Content of language	Lexicon	Vocabulary
	Semantics	Relationship among words. Symbols representing universal concepts. Meaning of words. (e.g., in relation to objects, agents, an action, states, attributes or locations). Meaning and relationship of abstract concepts (e.g., idioms and proverbs)
Use of language	Pragmatics	Rules or conventions for use of language in a socially and culturally appropriate manner and in the appropriate context, e.g., turn taking, eye contact, maintaining a topic in conversation

Used with permission from: Patel DR. Principles of developmental diagnosis. In: Greydanus DE, Feinberg A, Patel DR, *et al.* editors. Pediatric Diagnostic Examination. New York: McGraw Hill Medical, 2006:629-44.

recognized between 18 and 24 months of age when parents first bring their child concerned about his or her unusual behaviors or difficulty with social communication (5,12). A child with autism spectrum disorder may not be socially engaged, may demonstrate intense focus on a particular object for a prolonged period, may appear withdrawn at times, have poor eye contact, get attached to a particular object (toy), have specific rituals, and resist change in routine (5,12,32-38). A child with autism has difficulty understanding other person’s feelings, and may not engage in playing pretend games (5,12,22-38). Individuals with Asperger syndrome (also considered to be a form of high functioning autism spectrum disorder) demonstrate normal cognitive and language abilities and predominant deficits in social development (29-41).

A child with predominant cognitive deficit or intellectual disability generally has typical progression of motor development and his or her behaviors are commensurate with MA (18-20). Although a child with intellectual disability has both cognitive and adaptive functioning deficits, his or her social participation is also consistent with MA. In additional to language delay, a child with predominant cognitive deficit has difficulty problem solving and may not be able to relate their behaviors to consequences (18-20). In most cases, no specific etiology can be identified in children with intellectual disability, especially those with mild deficits; a specific cause is more likely to be identified in those with severe deficits (15,17-21). The most common cause of inherited intellectual disability is Fragile X syndrome (19,20). Other major causes of intellectual disability are fetal alcohol syndrome, lead toxicity, iron deficiency, and congenital brain malformations (15,17-21).

Developmental language disorder is characterized by predominant deficit in language development and typical development in other domains—social, motor and cognitive (24-26,30). In addition to autism and intellectual disability, the differential diagnosis of speech and language disorders should include hearing deficit, speech and voice disorders, maturational language delay, and a lack of environmental stimulation for language and learning (24-26). A bilingual home environment is not a cause for language delay. The various sub-types of developmental language disorders have been described that are based on particular aspects of the language that is affected (*Table 7*) (24-30).

Disorders of speech include speech sound disorder, stuttering, dysarthria, verbal dyspraxia and resonance disorders (*Table 8*) (5,25). Disturbance of airflow through nasopharyngeal airway results in either hypernasal or hyponasal voice; it is important to note that airflow obstruction is associated with the production of consonant sounds, and not associated with the production of vowel sounds (25). Selective mutism is not considered a language disorder, rather a specific type of anxiety disorder in which a child consistently fails to be able to speak in a given situation when he or she is expected to speak (5).

Regression of or failure to acquire expected new skills

In addition to developmental regression seen in children with autism, other conditions in which regression of skills is seen include Rett syndrome and Landau-Kleffner syndrome (LKS) (10,14). A child with Rett syndrome shows regression of acquired skills following a period of relatively typical development during infancy. Rett syndrome is an X-linked disorder predominantly affecting females (10). Other clinical features of Rett syndrome are autistic behavior, abnormal wringing hand movements, deceleration of head circumference, hyperventilation, breath holding, air swallowing, gait dyspraxia, autonomic dysfunction, inappropriate laughing, and intense eye contact (10). Often there is some recovery following a period of regression followed by further deterioration.

The clinical features of LKS are generally recognized between 3 and 8 years of age (10). Following a period of typical development during the first 3 years, the child with LKS loses language skills; whereas, no deficits are noted in cognitive or social development (10). The sleep electroencephalograph in LKS shows seizure pattern (10). Other causes of developmental regression in children older than 2 years of age include some genetic disorders, inborn errors of metabolism, and disorders of white and gray matter (10).

Early learning difficulties and behavioral symptoms

Learning difficulty and associated behavioral symptoms are most apparent first during early elementary grades. Poor academic performance is often the main parental concern. Other indications of learning difficulty include delay in completing assigned school work, inattentiveness in the classroom setting, difficulty learning new skills, and difficulty with reading and comprehension (42,43). Children with learning difficulty are also shy and reluctant to participate in activities with other children. In addition to various types of specific or developmental learning disorders other conditions to be considered in the differential diagnosis in these children include attention deficit hyperactivity disorder (ADHD), hearing or vision impairment, cognitive deficit and developmental coordination disorder (42).

Vision and hearing impairment may be associated with other developmental disabilities. A child with visual impairment might close or cover one eye; squint the eyes or frown; complaint that things are blurry or hard to see; have trouble reading or doing other close-up work, or hold objects close to eyes; blink more than usual or seem cranky when doing close-up work such as looking at a book. A child with complete or partial hearing impairment might not turn to the source of a sound from birth to 3 or 4 months of age; may not say single words, such as “dada” or “mama” by 1 year of age; turn head when he or she sees you but not if you only call out his or her name. This often is mistaken for not paying attention or just ignoring.

Developmental coordination disorder affects school-age children and persists into adolescent years (5). Difficulties in motor coordination will cause substantial impairment in academic function or activities of daily living (5). Early manifestations may include difficulty sucking and swallowing, drooling during infancy, speech difficulties and delayed motor milestones during early childhood. Parents may observe that the child has difficulties with many of the fine motor tasks such as using scissors, tying shoe laces or buttoning or unbuttoning (5). They also may drop objects, have poor handwriting, or will frequently bump into furniture or other people. Differential diagnosis in these children include ADHD, visual impairment, and intellectual disability.

Table 7 Sub-types of developmental language disorders

Disorder	Key characteristics
Verbal dyspraxia (developmental apraxia of speech)	Expressive language disorder
	Difficulty planning, sequencing and executing voluntary speech sounds
	Dysfluent, significantly delayed, unintelligible speech
	Inconsistent articulation errors
Speech programming deficit disorder	Expressive language disorder
	Fluent, unintelligible, jargon
	Considered by some to be similar to verbal dyspraxia
Verbal auditory agnosia (word deafness)	Mixed receptive-expressive disorder
	Profound impairment in comprehension of spoken words
	Most children non-verbal or very limited verbal expression
	Very rare in children
Phonologic or syntactic deficit disorder	Mixed receptive-expressive disorder
	Comprehension relatively better than expression
	Telegraphic speech, limited vocabulary
	Grammatical errors
	Significant omissions, distortions, and substitution of words
	Short sentences, difficulty in repeating words or sentences
Lexical deficit disorder (lexical-syntactic deficit)	Higher order processing disorder
	Severe deficits in word finding a paraphrasing
	Jargon and pseudo stuttering
	Significant deficiency in understanding connected speech
	Impoverished syntax, syntactic distortions
	Spontaneous language relatively better than language on demand
	Ability to decode wh- questions is limited
Semantic-pragmatic deficit disorder	Higher order processing disorder
	Poor discourse of connected speech
	Apparent talkativeness
	Phonology and (simplified) syntax preserved
	Atypical choices of words
	Word finding deficits
	Significant deficits in comprehension and verbal reasoning
	Tangential and stereotyped speech, echolalia
	Speaking aloud to no one in particular
	Poor maintenance of topic
	Responding inaccurately or out of context
	Rare disorder in children

Table 8 Developmental speech disorders

Disorder	Description
Speech sound disorder	Also described as functional articulation disorder or phonological disorder. Characterized by errors in articulation and speech sounds, consistent substitution of simple sounds for complex sounds or single consonants for blended consonants, dropped consonants, and errors within words. Problem may not be recognized until preschool
Stuttering	Disturbed speech fluency with atypical rate and rhythm, and repetitions of sounds, syllables, words, and phrases generally accompanied by evidence of stress or physical tension. There may be sound prolongations, interjections, pauses within words, blocking of words. Typical onset between 2 and 7 years with peak at age 5 years
Resonance disorders	Can be either hypernasal or hyponasal voice due to anatomical factors. Hypernasality may be due to dysfunction of the velopharyngeal mechanism, seen for example in cleft palate. Hyponasality is seen, for example, in nasal congestion, upper respiratory infections, nasal anomalies, hypertrophied adenoids
Dysarthria	Due to dysfunction of the neuromuscular or motor mechanism for speech production (e.g., cerebral palsy). Characterized mainly by inconsistent misarticulations of speech sounds and words, poor intelligibility, and slow speech
Verbal dyspraxia and speech programming disorder	Both terms describe similar types of largely speech production problems. These disorders may significantly influence expressive language as well

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Adolescents

Academic difficulties

The predominant concern in adolescents is difficulty in learning as the burden and complexity of academic work increase with advancing grade levels. Difficulty in learning may affect one or more areas of learning. Children and adolescents with specific learning disorder may present with associated behavioral symptoms. The differential diagnosis of learning disorders should include specific or developmental learning disorder, anxiety, depression, ADHD, and substance use disorders (42-57). Learning is also affected by the adequacy and quality of support system and instructions. Nutritional and psychosocial factors can also affect learning and academic performance.

Specific or developmental learning disorder is diagnosed when scores on an individually administered standardized achievement tests (in reading, mathematics, or written expression) are substantially below that expected for the child's age, intelligence and education level (5,43). Non-verbal specific learning disorder is characterized by difficulty in problem solving ability, and deficits in visual-spatial and visual-perceptual abilities; language-based skills and intelligence are normal. Key features of reading disorder, mathematics disorder and disorder of written expression are summarized in *Table 9* (42-58).

Clinical approach

Surveillance

Developmental and behavioral surveillance is recommended at every preventive or health maintenance visit (1,11). Information should be obtained from multiple sources including care givers and parents, teachers, and academic performance reports. A detailed prenatal, birth, perinatal, neonatal, and developmental history is essential aspect in the diagnostic evaluation of children with atypical development. Parental concerns regarding child's development and behavior provides a basis for the direction of evaluation. The developmental history should assess areas of typical and atypical progression of milestones in all streams or domains of development—gross motor, fine motor, speech and language, and social (1).

Although, specific emphasis on a particular aspect of physical examination is guided by the presenting symptoms, some aspects are of fundamental consideration in all children. These include serial measurements of growth parameters—height, weight, head circumference, visual acuity, hearing, and a thorough search for congenital anomalies and dysmorphic features. The growth parameters should be plotted on appropriate graphs. In older children and adolescents mental status examination should be included and in adolescents external genital examination

Table 9 Key features of specific learning disorders

Disorder	Most likely to be recognized by	Some key features
Mathematics	2nd or 3rd grade level	Key problem is with arithmetics
		Difficulties in:
		Understanding or naming mathematical terms
		Math operations or concepts
		Decoding or recognizing math symbols or signs
		Copying numbers or figures
		Following sequences of math steps
		Counting or multiplication tables
Reading	4th grade level	Delayed language milestones
		Spelling errors
		Slow reading
		Problems with rhyming words or words that sound alike
		Difficulty decoding unfamiliar or nonsense words or single words
Written expression	5th grade level	Poor writing skills
		Grammatical errors
		Punctuation errors
		Poor paragraph organization
		Spelling errors
		Very poor handwriting

sexual maturity rating should be noted.

The pattern of atypical progression of development—delay, dissociation, deviation, regression—is ascertained based on the information obtained from the history and physical examination. Developmental maturation and behavior of the child are closely linked and any evaluation of child's development should include a behavioral history. Child's interactions and behavior, both in familiar (home, school) and unfamiliar (community) settings, should be ascertained. Information should be obtained about child's interactions with peers, parents, other care givers, teachers, and other adults (1). The behavioral history provides the basis to describe any atypical pattern of child's behavior, such as aggression, impulsivity, decreased social engagement, repetitive behaviors or self-injurious behavior (1). Behavior is generally assessed based on direct observation, spontaneous activities or responsive behaviors across different settings (1). As part of surveillance, information should be obtained about possible risk and protective factors relative to child's developmental and behavioral concerns (1,3,11).

Screening

Developmental screening that screen across all streams of development with appropriate standardized screening instrument is recommended as part of the periodic health maintenance examinations in the primary care setting at 9, 18, 24 or 30 months of age (*Table 10*) (1,11). Although screening is across all streams, certain areas are of specific focus at each visit. In addition, autism specific screening is recommended at 18- and 24-month visits (1,11,12). Examples of standardized tools for general developmental screening, general behavioral screening, language screening, and autism screening are listed in *Table 11* (59-67).

It is important to use standardized, validated screening tool to reduce the false positive and false negative results, which can have implications for further evaluation. A systematic review analyzed 11 studies that described developmental screens performed by primary care physicians (6). This review determined the sensitivity of these screens to range between 14% and 54% while their specificities ranged between 69% and 100%. There was 1 outlier in this review with a sensitivity of 85% and a specificity of 61%. Sensitivity of the developmental screenings, excluding the outlier, was consistently low. Thus, developmental screens correctly identified children with developmental delays a low number of times. On the other hand, the specificity of the developmental screens was up to the standards of 70%.

Table 10 Screening

Screening	Focus area	Identified concerns
9 months	Vision	Hearing deficit
	Hearing	Vision deficit
	Gross and fine motor	Neuromotor problem
	Receptive language	
18 months	Expressive language	Hearing and vision deficits
	Receptive language	Autism
		Language problem
		Cognitive deficits
30 months	Behavioral interactions	Attention problems
		Disruptive behaviors

Evaluation

A more comprehensive evaluation is guided by findings on developmental surveillance and screening. Specific clinical psychological or neuropsychological evaluation and psychological testing should be performed by clinical psychologists with appropriate expertise and experience in the evaluation of children with atypical development. Based on findings on clinical history and physical examination further medical evaluation may be indicated. Such evaluation may include neuroimaging, electroencephalography, tests for genetic disorders, and specific laboratory tests for inborn errors of metabolism and should be considered in consultation with appropriate medical specialists (11,12,14,15).

Both hearing screening and vision screening are part of periodic health maintenance or preventive health visits. In addition, parental concerns about a child's hearing or vision are indications for further audiological testing and complete vision evaluation. In the United States, based on current guidelines, universal hearing screening is done in all newborns by 1 month of age, with a follow-up audiological testing, if indicated, by 3 months of age. Treatment

Table 11 Examples of standardized screening tools

Category	Screening tool	Source
General developmental screening	Ages and States Questionnaires-3 (ASQ-3)	Paul H Brooks Publishing
		www.brookespublishing.com
	Parents' Evaluation of Developmental Status (PEDS)	Ellsworth & Vandermeer Press; www.pedstest.com
	Parents' Evaluation of Developmental Status: Developmental Milestones (PEDS:DM) Screening Version	
	Survey of Wellbeing of Young Children (SWYC)	www.theswyc.org
General behavioral screening	Ages and Stages Questionnaire: Social-Emotional-2 (ASQ:SE-2)	www.agesand stages.com
	Brief Infant Toddler Social Emotional Assessment (BITSEA)	Pearson Assessments
		https://www.pearsonassessments.com
	Pediatric Symptom Checklist-17 items (PSC-17b)	https://www.massgeneral.org
Language	Communication and Symbolic Behavior Scales: Developmental Profile (CSBS-DP): Infant Toddler Checklist	www.brookespublishing.com
Autism	Modified Checklist for Autism in Toddlers. Revised with Follow-up (M-CHAT-R/F)	www.m-chat.org
	Social Communication Questionnaire (SCQ)	www.wpspublish.com

intervention is started by 6 months of age in any infant identified having a hearing loss. Similarly, any concerns about a child's speech and language development is an indication for complete evaluation by a speech-language pathologist. Often, no specific signs or symptoms are evident suggestive of a developmental disorder; however, based on their observations, parents remain concerned about their child's development. In these cases, a further evaluation is indicated.

Principles of management

The key principle of management of children with developmental concerns is early intervention. The content and delivery of early intervention services may vary in different health systems. For example, in the United States, many federal and state regulations provide a basis and framework for such services to be delivered within the context of the local formal education system (68,69). When a developmental problem is recognized, the child is enrolled in the early intervention service, while further evaluation may proceed at the same time. This approach is based on the principles of response to intervention concept (68,69). Rather than waiting for a definitive diagnosis, intervention is implemented and further course of action is guided by response to the intervention. Among the major services included in the intervention programs are occupational therapy, physiotherapy and speech therapy. School age children are evaluated further to develop individualized learning plans that include various remediation, accommodation or psychoeducational strategies (43). In addition, environmental risk factors are considered and addressed in home, school and community settings as appropriate. When a specific developmental or behavioral disorder is identified, treatment is tailored to the condition and may include specific behavioral and pharmacological interventions.

Conclusions

The development from infancy through adolescence is described under four main streams or domains of development, namely, (I) gross and fine motor, (II) speech-language, (III) social-emotional, and (IV) cognitive. However, overall typical development progresses concomitantly in all domains. Typical development follows certain well recognized patterns, an understanding of which, helps us identify atypical patterns of developmental progression. Developmental delay, deviation, dissociation and regression are recognized as atypical patterns of development. The predominant clinical presentations of atypical patterns of development vary depending up on the age of the infant or child at presentation. The DQ is used as a measure of rate of development in any given domain; similarly, intelligence quotient is used as measure of intelligence. The clinical approach to developmental diagnosis is based on the principles of regular surveillance, periodic screening and diagnostic evaluation when indicated based on finding on surveillance and screening. Early recognition of atypical development is important as early treatment interventions, based on response to intervention model, have been shown to the most effective.

Acknowledgments

Funding: None.

Footnote

Conflicts of Interest: DRP serves as the unpaid Deputy Editor-in-Chief of *TP* and the unpaid Guest Editor of the focused issue "Neurodevelopmental and Neurobehavioral Disorders in Children". *TP*. Vol 9, Supplement 1 (February 2020). Section on clinical presentations is adapted with permission from author's previous work: Patel DR. Principles of developmental diagnosis. In: Greydanus DE, Feinberg A, Patel DR, *et al.* editors. Pediatric Diagnostic Examination. New York: McGraw Hill Medical, 2006:629-44. The other authors have no conflicts of interest to declare.

Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Childhood (Key Ideas) | Jenks

Week 1, Class 2

Chapter 1: Constituting Childhood (selection) — pp. 1–8
Developmental Psychology — The Piagetian Paradigm — pp. 22–30

Struck-through text on pages 8, 22, and 30 falls outside the assigned selection.

Jenks, Chris. 1996. Childhood. Key Ideas. Routledge. Pages 1–8; 22–30.

Constituting childhood

We would not have our Guardians grow up among representations of moral deformity, as in some foul pasture where, day after day, feeding on every poisonous weed they would, little by little, gather insensibly a mass of corruption in their very souls. Rather we must seek out those craftsmen whose instinct guides them to whatsoever is lovely and gracious; so that our young men, dwelling in a wholesome climate, may drink in good from every quarter, whence, like a breeze bearing health from happy regions, some influence from noble works constantly falls upon eye and ear from childhood upward, and imperceptibly draws them into sympathy and harmony with the beauty of reason, whose impress they take. Hence . . . the decisive importance of education in poetry and music; rhythm and harmony sink deep into the recesses of the soul and take the strongest hold there, bringing that grace of body and mind which is only to be found in one who is brought up in the right way. Moreover, a proper training in this kind makes a man quick to perceive any defects or ugliness in art or in nature. Such deformity will rightly disgust him. Approving all that is lovely, he will welcome it home with joy into his soul and, nourished thereby, grow into a man of noble spirit. All that is ugly and disgraceful he will rightly condemn and abhor while he is still too young to understand the reason; and when reason comes, he will greet her as a friend with whom his education has made him long familiar.

Plato, *The Republic*

In what ways can we possibly begin to make sense of children? This is by no means a novel question. Since humankind fir t achieved parenthood the problem has stalked the adult condition. But at a different level we might rightly suppose that from the earliest Socratic dialogue onwards through the history of ideas, moral, social and political theorists have systematically endeavoured to constitute a view of the child that is compatible with their particular visions of social life and continuous with their speculations concerning the future. Beginning from that initial Hellenic desire to seek out the origins of virtue in order to instil rhythm and harmony into the very souls of the young, and extending up until our contemporary pragmatic concerns with the efficacy of specific and fashionable child-rearing practices, after centuries of debate and practice, we have still not achieved any consensus over the issue of childhood. Despite a long cultural commitment to the good of the child and a more recent intellectual engagement with the topic of childhood, what remains perpetually diffuse and ambiguous is the basic conceptualization of childhood as a social practice. Childhood remains largely unrealized as an emergent patterning of action. As Rousseau stated in the Preface to *Émile*:

> We know nothing of childhood: and with our mistaken notions the further we advance the further we go astray. The wisest writers devote themselves to what a man ought to know, without asking what a child is capable of learning. They are always looking for the man in the child, without considering what he is before he becomes a man.

What do we bring to mind when we contemplate the child? Whether to regard children as pure, bestial, innocent, corrupt, charged with potential, *tabula rasa*, or even as we view our adult selves; whether they think and reason as we do, are immersed in a receding tide of inadequacy, or are possessors of a clarity of vision which we have through experience lost; whether their forms of language, games and conventions are alternatives to our own, imitations or crude precursors of our own now outgrown, or simply transitory impenetrable trivia which are amusing to witness and recollect; whether they are constrained and we have achieved freedom, or we have assumed constraint and they are truly free – all these considerations, and more, continue to exercise our theorizing about the child in social life.

Any review of the multiplicity of perspectives that are emerging in relation to childhood and also those that have previously been adopted in this area of study reveal, at one level, a continuous paradox, albeit expressed in a variety of forms. Simply stated the child is familiar to us and yet strange, he or she inhabits our world and yet seems to answer to another, he or she is essentially of ourselves and yet appears to display a systematically different order of being. The child's serious purpose and our intentions towards him or her are dedicated to a resolution of that initial paradox by transforming him or her into an adult, like ourselves. The reflexive recognition of this sustained, and sustaining, commitment opens up the whole related set of questions concerning the necessity and contingency of the relationship between the child and the adult, both in theory and in everyday life. The 'known' difference between these two social locations directs us towards an understanding of the identity contained within each; the contents are marked by the boundaries. The child, therefore, cannot be imagined except in relation to a conception of the adult, but essentially it becomes impossible to generate a well-defined sense of the adult, and indeed adult society, without first positing the child. The relationship child–adult appears locked within the binary reasoning which, for so long, both contained and constrained critical thought in relation to issues of gender and ethnicity. The child, it would seem, has not escaped or deconstructed into the post-structuralist space of multiple and self-presentational identity sets.

From this formulation we may distil two elements that appear common to the mainstream of approaches to the study of childhood. First, a foundational belief that the child instances difference and particularity (a belief that we shall later explore), and second, following from the former, a universal cultural desire to both achieve and account for the integration of that difference into a more broadly conceived sense of order and generality that comprises adult society. This is an integration predicted and condemned by Rousseau:

> Nature wants children to be children before they are men. If we deliberately pervert this order, we shall get premature fruits which are neither ripe nor well-flavoured, and which soon decay. . . . Childhood has ways of seeing, thinking, and feeling peculiar to itself; nothing can be more foolish than to substitute our ways for them.

Typically, how ver, the overwhelming irony of this manner of thinking is that it signally fails to attend to, or even acknowledge, its own par doxical character. Inevitably the child side of the relationship w thin analysis of this genre is uniformly recovered in a negative ashion. Such theories, which tend to be hegemonic within their specific disciplines, such as socialization theory and developmental psychology, are predicated upon a strong but unexplicated know edge of the difference of childhood and they proceed rapidly to an overattentive elaboration of the compulsive processes of integration. It is as if the basic ontological questions, 'What is a child?' nd 'How is the child possible as such?', were, so to speak, satisf ctorily answered in advance of the theorizing and then summar y dismissed.

THE CHILD AS 'SAVAGE'

In many ways this constitutes a reprise of nineteenth-century social thought. Just as the early 'evolutionist' anthropologist, a self-styled civilized person, simply 'knew' the savage to be different to himself, on a scale of advancement, and thus worthy of study; so we also, as rational adults, recognize the child as different, less developed, and in need of explanation. Both of these positions proceed from a pre-established but tacit ontological theory, a theory of what makes up the being of the other, be it savage or child. It is these unspoken forms of knowledge, these tacit commitments to difference that routinely give rise to the accepted definition of the savage or the child as a 'natural' meaningful order of object. Such implicit theories serve to render the child–adult continuum as utterly conventional and thus taken for granted for the modern social theorist as the distinction between primitive thought and rational thought was for the early anthropologist. Such social hierarchies are taken for granted in our cognitions because we do not examine the assumptions on which they are based. These assumptions embody the values and interests of the theorist and the contemporary culture, which in turn generate normative expectations within the wider society. However, the history of the social sciences has attested to a sequential critical address and debunking of the dominant ideologies of capitalism in relation to social class, colonialism in relation to race, and patriarchy in relation to gender; but as yet the ideology of development in relation to childhood has remained relatively intact. The politics of this situation warrant further exploration; its consequences are that the child, like the savage of an earlier epoch, is either excluded from our analysis or reimported as an afterthought. A major concern of this book is to encourage the reader to make a critical reconstruction of such sets of assumptions as they may be available, and to different degrees, in the literature concerning childhood that we shall explore together. In this way the child might by reinvented or at least recovered positively. Cunningham contributes to this question when he states that: 'Analogies between children and savages do not exist in a social or political vacuum.' He seeks to relate such analogies precisely to their contexts or, to use his own terms 'to identify discourses about childhood, and the power relationships which they embody.'[1]

At risk of exhausting the previously developed analogy it may be suggested that whereas the early anthropologist had to voyage to his chosen phenomena, like an explorer, across social space, the child as a contemporary topic has been most vividly brought into recognition through the passage of social time. Both of these journeys symbolize a questing after control through understanding but neither are appropriately exclusive in as much as both the synchronic and diachronic dimensions are pertinent to our knowledge of any socio-cultural form. What we should recognize however, is that both such processes are significant in fashioning their object; that is to say, comparative and historical analyses, in different ways, succeeded in establishing different classifications and boundaries around their phenomena. The manner of our assembling and the character of our distancing are significant in the constitution of either savage or child. A range of histories of childhood that we shall examine later demonstrate the erratic evolution of the image of childhood and its changing modes of recognition and reception. We will note that the child emerges in contemporary European culture as a formal category and as a social status embedded in programmes of care, routines of surveillance and schemes of education and assessment. Such accounts ensure that the child is realized as the social construction of a particular historical context and this provides a major platform for much contemporary theorizing about childhood; however it is the child's identity as a social status that etermines its difference and recognition in the everyday world. The status of childhood has its boundaries maintained through the c tallization of conventions and discourses into lasting institution l forms like families, nurseries, schools and clinics, all agencies specifically designed and established to process the child as a uniform entity. Comparative material drawn from cross-cultural contexts reveals divergent sets of conventions and discourses, and thus institutional forms, some utterly different from our own but others bearing strong resemblances, all are bound together through homology. The comparative material, as we shall see, instructs us to think more profitably of childhood(s) rather than of a singular and mono-dimensional status.

In the same way that the 'savage' served as the anthropologist's referent for humankind's elementary forms of organization and primitive classifications, thus providing a speculative sense of the primal condition of human being within the socio-cultural process, so also the child is taken to display for adults their own state of once untutored difference, but in a more collapsed form: a spectrum reduced from 'human history' to one of generations. It will be suggested that within the child humanity sees its immediate past but also contemplates the immortality of its immanent future.

In the everyday world the category of childhood is a totalizing concept, it concretely describes a community that at some time has everybody as its member. This is a community which is therefore relatively stable and wholly predictable in its structure but by definition only fleeting in its particular membership. Beyond this the category signifies a primary experience in the existential biography of each individual and thus inescapably derives its common-sense meanings, relevance and relation not only from what it might currently be as a social status but also from how each and every individual, at some time, must have been. It is the only truly common experience of being human, infant mortality is no disqualification.

THE 'NATURAL' CHILD

Perhaps because of this seemingly all-encompassing character of the phenomenon as a social status and because of the essentially personal character of its particular articulation, common-sense thinking and everyday language in contemporary society are rife with notions concerning childhood. Being a child, having been a child, having children and having continuously to relate to children are all experiences which contrive to render the category as 'normal' and readily transform our attribution of it to the realm of the 'natural' (as used to be the case with sex and race). Such understandings, within the collective awareness, are organized around the single most compelling metaphor of contemporary culture, that of 'growth'. Stemming from this, the physical signs of anatomical change that accompany childhood are taken to be indicators of a social transition, so that the conflation of the realms of the 'natural' and the 'social' is perpetually reinforced.

All contemporary approaches to the study of childhood are clearly committed to the view that childhood is not a natural phenomenon and cannot properly be understood as such. The social transformation from child to adult does not follow directly from physical growth and the recognition of children by adults, and vice versa, is not singularly contingent upon physical difference. Furthermore, physical morphology may constitute a form of difference between people in certain circumstances but it is not an adequately intelligible basis for the relationship between the adult and the child. Childhood is to be understood as a social construct, it makes reference to a social status delineated by boundaries that vary through time and from society to society but which are incorporated within the social structure and thus manifested through and formative of certain typical forms of conduct. Childhood then always relates to a particular cultural setting.

Our early anthropologist would readily recognize the significance of such concepts within the social life of his 'savage'; he would demonstrate that the variety and hierarchy of social statuses within the tribe are plainly prescribed by boundaries which are, in turn, maintained through conventional practices deeply bound within ritual. Any transposition from one status location to another is never simply a matter of physical growth or indeed physical change. Such movements require transformatory process such as valediction, rites of passage and initiation ceremonies, all of which are disruptive and painful and have an impact not just upon the individual but also upon the collectivity. The recognition that we are addressing the somewhat more diffuse and volatile boundaries that mark off childhood today, and the fact that we are considering such a transition from within the mores and folkways of a modern secular society, is no guarantee that the ritualism will be any less present. Rather, perhaps, the rituals will have become more deep-seated and ideological in their justification. Whatever, the experience of change through ritual will continue to exercise a violent and turbulent constraint on the individual's consciousness. We might argue, in fact, that since the turn of the twentieth century we have developed a psychoanalytic vocabulary of motives that ascribe all pathological conduct to the dysfunctional integration of the effects of the culturally based rituals that are instrumental in our becoming adult.

The widespread tendency to routinize and 'naturalize' childhood, both in common sense and in theory, serves to conceal its analytic importance behind a cloak of the mundane; its significance and 'strangeness' as a social phenomenon is obscured. Within everyday rhetoric and many discourses of theory childhood is taken for granted, it is regarded as necessary and inevitable, and thus part of normal life – its utter 'thereness' seems to foster a complacent attitude. This naturalism has, up until fairly recently, extended to the social sciences, particularly psychology, where childhood is apprehended largely in terms of biological and cognitive development through concepts such as 'maturation'. Sociology, in search of explanations through structural causality, has independently sought to understand the problem of the child's acquisition of specific cultural repertoires through the largely one-sided theories of socialization. All of these ways of proceeding, though predominant within the academy, leave the actual child untheorized, they all contrive to gloss over the social experience that is childhood. We might here concur with Hillman who, albeit writing in the context of pastoral psychology, states that much of what is said about children and childhood is not really about children and childhood at all.

> What is this 'child' – that is surely the first question. Whatever we say about children and childhood is not altogether about children and childhood. What is this peculiar realm we call 'childhood' and what are we doing by establishing a special world with children's rooms and children's toys, children's clothes, and children's books. Clearly, some realm of the psyche called 'childhood' is being personified by the child and carried by the child for the adult.[2]

DEVELOPMENTAL PSYCHOLOGY – THE PIAGETIAN PARADIGM

Perhaps the irony of the exclusion of the child through partial formulations of rationality is nowhere more fundamentally encountered than in the body of work known as developmental psychology. This has been defined by Piaget as follows:

> Developmental psychology can be described as the study of the development of mental functions, in as much as this development can provide an explanation, or at least a complete description, of their mechanisms in the finished state. In other words, developmental psychology consists of making use of child psychology in order to find the solution to general psychological problems.[17]

However, as Burman has pointed out:

> Nowadays the status of developmental psychology is not clear. Some say that it is a perspective or an approach to investigating general psychological problems, rather than a particular domain or sub-discipline. According to this view we can address all major areas of psychology, such as memory, cognition, etc., from this perspective. The unit of development under investigation is also variable. We could be concerned with the development of a process, or a mechanism, rather than an individual. This is in marked contrast with the popular representations of developmental psychology which equate it with the practicalities of child development or, more recently, human development.[18]

Leading within this field, and heading the 'popular representations', is the work of Piaget and his theories of intelligence and child development which have had a global impact on paediatric care and practice. Piaget's 'genetic epistemology' seeks to provide a description of the structuring of thought and finally the rational principle of nature itself, all through a theory of learning. As such Piaget's overall project represents a significant contribution to philosophy as well. Following within the neo-Kantian tradition his ideas endeavour to conciliate the divergent epistemologies of empiricism and rationalism; the former conceiving of reality as being available in the form of synthetic truths discoverable through direct experience, and the latter viewing reality analytically through the action of pure reason alone. Kant, in his time, had transcended this dichotomy through the invocation of 'synthetic *a priori* truths' that are the immanent conditions of understanding, not simply amenable to logical analysis. Piaget's categories of understanding in his scheme of conceptual development may be treated as being of the same order. His work meticulously constitutes a particular system of scientific rationality and presents it as being both natural and universal. However, as Archard stated:

> Piaget suggested that all children acquire cognitive competencies according to a universal sequence. Nevertheless, he has been criticised on two grounds. . . . First, his ideal of adult cognitive competence is a peculiarly Western philosophical one. The goal of cognitive development is an ability to think about the world with the concepts and principles of Western logic. In particular Piaget was concerned to understand how the adult human comes to acquire the Kantian categories of space, time and causality. If adult cognitive competence is conceived in this way then there is no reason to think it conforms to the everyday abilities of even Western adults. Second, children arguably possess some crucial competencies long before Piaget says they do.[19]

Piaget's empirical studies concerning the development of thought and intelligence describe, what are for him, the inevitable and clearly defined stages of intellectual growth that begin from *sensory-motor* intelligence immediately succeeding birth, and proceed through *pre-conceptual* thought, *intuitive* thought, and *concrete* operations up to the level of *formal operations*, for most people, in early adolescence. These stages are chronologically ordered but also hierarchically arranged along a continuum from low status, infantile, 'figurative' thought to high status, adult, 'operative' intelligence. This sets a narrative in the discourse of cognitive growth that is by now global and overwhelming. The 'figurative' thought that emits directly from our state of childhood, is instanced by particularistic activity, a concentration on the here and now, and a consequent inability to transfer experience or training from one situation to another. The child, for Piaget, is preoccupied with the repetitive and highly concrete replication of object states, it is clearly dominated by objective structures and inhabits a material universe. Beyond this figurative thought knows no distance or consideration, it is organized through affective responses in specific settings. Clearly we have here a recipe for 'childish-ness'.

Operative intelligence, on the other hand, the magnetic conclusion to the story, is and ought to be the province of adulthood. It implies the informed cognitive manipulation and transformation of objects by a reflecting subject. Operative intelligence is ideal, it exemplifies logical process and freedom from domination by immediate experience.

Within Piaget's system each stage of intellectual growth is characterized by a specific 'schema' or well-defined pattern and sequence of physical and mental actions governing the child's orientation to the world. Thus the system has a rhythm and a calendar too. The development and transition from figurative to operative thought, through a sequence of stages, contains an achievement ethic. That is to say that the sequencing depends upon the child's mastery and transcendence of the schemata at each stage. This implies a change in the child's relation to the world. This transition, the compulsive passage through schemata, is what Piaget refers to as a 'decentring'. The decentring of the child demonstrates a cumulative series of transformations: a change from solipsistic subjectivism to a realistic objectivity; a change from affective response to cognitive evaluation; and a movement from the disparate realm of value to the absolute realm of fact. The successful outcome of this developmental process is latterly typified and celebrated as 'scientific rationality'. This is the stage at which the child, now adult, becomes at one with the logical structure of the cosmos. At this point, where the child's matured thought provides membership of the 'circle of science' the project of 'genetic epistemology' has reached its fruition, it is complete.

However, as Venn and Walkerdine pointed out:

> For Piaget, the individual subject is an exemplar, the typical representative of the species. He subscribes to the Lamarckian idea of cumulative assimilation, whereby the characteristics of individuals over time are resorbed into a single intellectual organism. Thus the processes, including those of cognitive development are the same in all single individuals, so that one need only study any exemplar and generalise.[20]

Concretely, scientific rationality for Piaget is displayed through abstraction, generalization, logico-deductive process, mathematization and cognitive operations. At the analytic level, however, this rationality reveals the intentional character of Piaget's theorizing and grounds his system in the same manner as did Parsons' transcendent 'cultural values'. Furthermore, whereas socialization through identification provided the key to the dissolution of the child within the social system, so also within Piaget's genetic epistemology, the process can be exposed as the analytic device by and through which the child is wrenched from the possibility of difference within the realm of value and integrated into the consensus that comprises the tyrannical realm of fact. Scientific rationality or adult intelligence is thus the recognition of difference grounded in unquestioned collectivity – we are returned to the irony contained within the original ontological question. The child is, once more, abandoned in theory. Real historically located children are subjected to the violence of a contemporary mode of scientific rationality which reproduces itself, at the expense of their difference, beyond the context of situated social life. Venn and Walkerdine commented further that:

> For Piaget the development of thought becomes 'indifferent to' the actual content of thought and the material base, although constructed out of it. He speaks of action on the concrete as being the basis out of which the operational structures are constructed, but his account is in the end unsatisfactory because he is more concerned with the results of abstraction as indicators of the way the mind works.[21]

The 'fact' of natural process overcomes the 'value' of real social worlds. And the normality of actual children becomes scrutinized in terms of the norms predicted by developmental psychology.

Rose, commenting on the historical context of this oppressive tendency, stated:

> Developmental psychology was made possible by the clinic and the nursery school. Such institutions had a vital role, for they enabled the observation of numbers of children of the same age, and of children of a number of different ages, by skilled psychological experts under controlled experimental, almost laboratory, conditions. Thus they simultaneously allowed for standardization and normalization – the collection of comparable information on a large number of subjects and its analysis in such a way as to construct norms. A developmental norm was a standard based upon the average abilities or performances of children of a certain age in a particular task or a specified activity. It thus not only presented a picture of what was *normal* for children of such an age, but also enabled the normality of any child to be assessed by comparison with this norm.[22]

Piaget's developmental theory states that the dynamic of the decentring process is provided by the interplay between two fundamental, complementary processes, being 'assimilation' and 'accommodation'. The two processes concern the child's choice and relation to that which is other than himself: assimilation concerns the absorption and integration of new object experiences into existing and previously organized schemata, whereas accommodation involves the modification of existing schemata, or the construction of new schemata to encapsulate new and discordant object experiences. These two processes are complementary in that accommodation generates new organizing principles with which to overcome the 'disequilibrium' produced by the new experiences that cannot be readily assimilated. Within Piaget's demonstrations of adult scientific rationality, the child is deemed to have appropriately adapted to the environment when he or she has achieved a balance between accommodation and assimilation. It would seem that the juggling with homeostasis is forever the child's burden! However, although from a critical analytic stance accommodation might be regarded as the source of the child's integration into the consensus reality, within the parameters of the original theory the process is treated as the locus of creativity and innovation – it is that aspect of the structuring of thought and being which is to be most highly valued. In contradistinction Piaget regards children's play as non-serious, trivial activity in as much as it displays an emphasis on assimilation over accommodation. Play is merely diverting fun or fantasy, it deflects the child from his true destiny and logical purpose within the scheme of rationality. The problem is that the criteria for what constitutes play need not equate with the rigorous, factual, demands of reality. Treating play in this manner, that is from the perspective of the rational and 'serious' adult, Piaget is specifically undervaluing what might represent an important aspect of the expressive practices of the child and his or her world. Following the work of Denzin:

> Childhood is conventionally seen as a time of carefree, disorganised bliss. . . . The belief goes that they [children] enjoy non-serious, play-directed activities. They avoid work and serious activities at all costs. . . . There is a paradox in these assumptions.[23]

And Stone:

> Educators too often have a restricted view of play, exercise, and sport, asking only how such activities contribute to the motor efficiency and longevity of the organism. Yet the symbolic significance of recreation is enormous, providing a fundamental bond that ties the individual to his society. . . . Social psychologists have long recognised the significance of play for preparing young children to participate later on in adult society.[24]

In this context I would argue that play is indeed an important component of the child's work as a social member. And I would argue further that play is instrumental in what Speier[25] has designated the child's 'acquisition of interactional competencies'. Genetic epistemology wilfully disregards, or perhaps just pays insufficient attention to, play in its urge to mathematize and thus render formal the 'rational' cognitive practices of adult individuals in their collective lives.

By treating the growth process of the child's cognition as if it were impelled towards a prestated structure of adult rationality, Piaget is driven to concur with Lévi-Bruhl's concept of the 'primitive mentality' of the savage, but in this instance in relation to the 'pre-logical' thought of the child. A further consequence of Piaget's conceptualization of the rational development of the child's 'embryonic' mind as if it were a natural process, is that the critical part played by language in the articulation of mind and self is very much understated. Language is treated as a symbolic vehicle which carries thought and assists in the growth of concepts and a semiotic system, but it is not regarded as having a life in excess of these referential functions. Thus language, for Piaget, is insufficient in itself to bring about the mental operations which make concept formation possible. Language, then, helps in the selection, storage and retrieval of information but it does not bring about the co-ordination of mental operations. This level of organization is conceptualized as taking place above language and in the domain of action. This is slightly confusing until we realize that action, for Piaget, is not regarded as the performative conduct that generates social contexts, but rather as a sense of behaviour that is rationally governed within the *a priori* strictures of an idealist metaphysics. Language, for Piaget, itemizes the world and acts as a purely cognitive function. This is a position demonstrably confounded by Merleau-Ponty in his work on the existential and experiential generation and use of language by children – the classic example being the child's generation of a past tense in order to express the loss of uniqueness and total parental regard following the birth of a sibling; language here is not naming a state of affairs but expressing the emotion of jealousy. Merleau-Ponty's work serves to reunite the cognitive and the affective aspects of being which are so successfully sundered by Piaget. He stated:

> I pass to the fact that appeared to me to be worthy of mention . . . the relation that can be established between the development of intelligence (in particular, the acquisition of language) and the configuration of the individual's affective environment.[26]

At the outset of this chapter I commented on the absence of any consensus view of the child within social theory. This is, perhaps, no bad thing. I have attempted to explicate certain of the normative assumptions at the heart of 'socialization theory' and 'developmental psychology' which have held as the orthodoxy up until recent years and I might optimistically suggest that such conventional explanations have been successfully supplanted by feminist theories in relation to the family and what have come to be grouped as 'social constructionist' views of the child, possibly instigated by this author but subsequently titled and joined by significant company.[27] However, a spurious consensus is not necessarily a desirable goal. It has been my intention to show that it is the different manners in which theoretical commitments are grounded that gives rise to the diversity of views of childhood.

My critical appraisals of Parsons and Piaget were not random selections. They have been singled out for analysis because of the clarity and penetration of their work, but more significantly because of the immense influence that they have exercised on the social sciences in the areas of socialization theory and learning theory. Extending beyond this both theorists, but particularly Piaget, have had an immeasurable impact upon the everyday common-sense conceptualization of the child. They and the paradigms that they have established have, to a large extent, captured and monopolized the child in social theory and in so doing have exemplified the point of this chapter. The idea of childhood is not a natural but a social construct and as such its status is constituted in particular socially located forms of discourse. Whether the child is being considered in the common-sense world or in the disciplined world of specialisms, the meaningfulness of the child as a social being derives from its place and its purpose in theory. Social theory is not merely descriptive and certainly never disinterested. In the variety of approaches that social theory adopts in relation to the child there is an analytic gathering around my central theme in this chapter, namely, that the child is constituted purposively within such theory. That is, the child is assembled intentionally to serve the purposes of supporting and perpetuating the fundamental grounds of and versions of humankind, action, order, language and rationality within particular theories. We are thus presented with different 'theoretical' children who serve the different theoretical models of social life from which they spring. The point is a phenomenological one.

My recommendation remains then, that a sociology of childhood should arise from the constitutive practices that provide for the child and the child–adult relationship. Any potential theorist of childhood who wishes to engage in such an analysis, as I have attempted with 'socialization theory' and 'developmental psychology', should realize that they too are responsible for constituting the child, and that different images and representations of the child are occasioned by the different theoretic social worlds that we inhabit. In this way the passage of our theorizing will continue to emerge from the stenosis of the dominant 'natural' archetypes of childhood, being those of either the pathological or the schismatic. We need no longer abandon the child either to ignorance and secondary status or to radical difference and a bipartite world.

Let us now, in the following chapter, explore some of the constraints placed upon a sociological conception of the child.

NOTES

[1] H. Cunningham, *The Children of the Poor: Representations of Childhood since the Seventeenth Century*, Oxford: Blackwell, 1991, p. 6.
[2] J. Hillman, *Loose Ends*, Irving (Texas): Spring Publications, 1975, p. 8.
[3] M. Speier, 'The everyday world of the child' in J. Douglas (ed.), *Understanding Everyday Life*, London: Routledge & Kegan Paul, 1970, p. 188.
[4] J. O'Neill, 'Embodiment and child development: a phenomenological approach' in H. P. Dreitzel (ed.), *Recent Sociology No. 5*, New York: Collier Books, 1973, p. 65.
[5] B. Bernstein, 'On the classification and framing of educational knowledge' in M. F. D. Young (ed.), *Knowledge and Control*, London: Collier-Macmillan, 1972, p. 47.
[6] P. Coveney, *The Image of Childhood*, Harmondsworth: Penguin, 1967, p. 29.
[7] I. Opie and P. Opie, *The Lore and Language of Schoolchildren*, Oxford: Oxford University Press, 1959.
[8] See F. Elkin and G. Handel, *The Child and Society: The Process of Socialization*, New York: Random House, 1972; N. Denzin, *Childhood Socialization*, San Francisco: Jossey-Bass, 1977; D. Goslin (ed.), *Handbook of Socialization Theory and Research*, Chicago: Rand McNally, 1969; K. Danziger (ed.), *Readings in Child Socialization*, Oxford: Pergamon; A. Morrison and D. McIntyre, *Schools and Socialization*, Harmondsworth: Penguin, 1971; G. White, *Socialization*, London: Longman, 1977.
[9] D. Wrong, 'The oversocialized conception of man in modern sociology' in *American Sociological Review*, XXVI, April 1961, p. 190.
[10] T. Parsons, *The Social System*, London: Routledge & Kegan Paul, 1951, p. 207.
[11] J. O'Neill, *The Poverty of Postmodernism*, London: Routledge, 1994, pp. 26–27.
[12] T. Parsons, op. cit., p. 205.